



Powers and roots using a calculator

Task A: Evaluate a number using a square root

1) **Evaluate** the following. If your answer is a decimal, write all the digits from your calculator display.

a) **Evaluate** $3.45^2 =$

b) **Evaluate** $1.2^2 + 13.4^2 =$

c) **Evaluate** $\sqrt{76.7376} - 2.3^2 =$

d) **Evaluate** $\sqrt{9^2 + 12^2} =$

2) Using a calculator, work out the missing lengths of the following shaded areas. Each shape is a square.



Area = 127.69 cm²



Area = 4.6225 m²



Area = 0.00 000 1 m²

3) Using a calculator, explain why the following evaluate to the answer given.

a) $\sqrt{\sqrt{\sqrt{65\,536}}} = 4$

b) $\sqrt{53^2} = 53$

Task B: Evaluate using exponents

1) **Evaluate** the following. If your answer is a decimal, write all the digits from your calculator display.

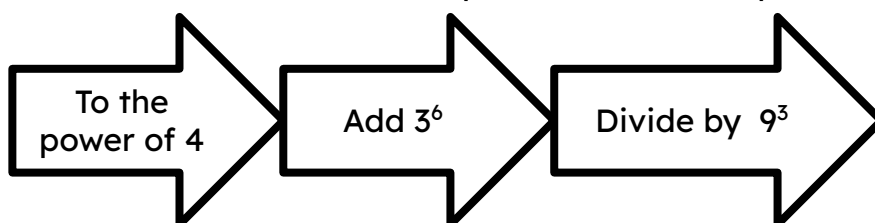
a) **Evaluate** $4^6 - 6^4 =$

b) **Evaluate** $27^4 \div 9^2 \div 3^7 =$

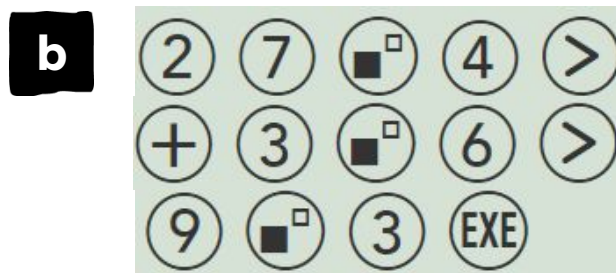
c) **Evaluate** $4.2^3 \div 2^4 =$

d) **Evaluate** $\sqrt{3^4 + 4^3 + 5^2 - 1^7} =$

2) Sofia followed this function machine. Her input was 27. Her output was 730.



Which of the following buttons did she press to get the correct answer?



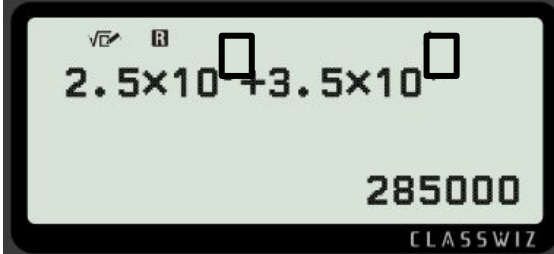
Task C: $\times 10^x$ button on a calculator1) **Evaluate** the following:

a) $1.7 \times 10^4 =$

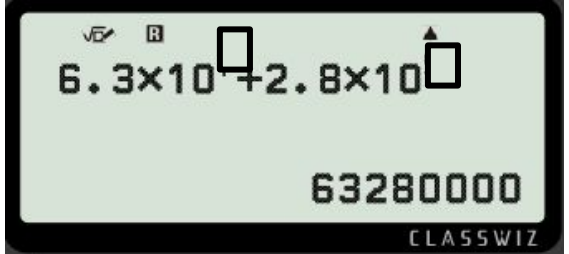
b) $3.9 \times 10^8 - 2.6 \times 10^7 =$

c) $1.2 \times 10^2 + 1.2 \times 10^3 + 1.2 \times 10^4 =$

2) Work out the missing exponents.

a) 

The calculator screen shows the expression $2.5 \times 10^{\square} + 3.5 \times 10^{\square}$ and the result 285000. The brand name CLASSWIZ is visible at the bottom.

b) 

The calculator screen shows the expression $6.3 \times 10^{\square} + 2.8 \times 10^{\square}$ and the result 63280000. The brand name CLASSWIZ is visible at the bottom.

c) 

The calculator screen shows the expression $3.1 \times 10^{\square} - 2.7 \times 10^{\square}$ and the result 3099730000. The brand name CLASSWIZ is visible at the bottom.